

Detecting departure and destination aerodromes from ADS-B

Version 6.1 — the altitude cut, and which datum it is measured against

EUROCONTROL Performance Review Unit

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1 What this version changes

Version 6 tuned `trend`'s altitude cut and left its *datum* alone. This version changes the datum, and measures what that is worth.

`trend` admitted a state vector to its vote when the aircraft was below a **flight level** — pressure altitude above the standard datum, the same number everywhere. `endpoint` had never worked that way: since version 6 its height gate has been measured **above the aerodrome's own field elevation**, on the stated grounds that “a fixed altitude cut-off means nothing at an aerodrome sitting at 5,000 ft”. Step 04's ground-phase detection moved to the same datum for the same reason.

`trend` was the last of the three still referenced to sea level. That is the asymmetry this version removes.

The consequence is not inaccuracy but **silence**, and it falls unevenly. At FL60 an aerodrome at sea level offers 6,000 ft of climb and descent for the vote to read. Madrid, at 1,998 ft, offers about 4,100. Ankara, at 3,125 ft, about 2,975. The altitude trace has progressively less room to rise or fall *clearly*, so the method abstains more often — and it abstains in proportion to how high the aerodrome sits.

i What this change cannot do

trend votes on the **sign** of a smoothed altitude change. Adding a constant offset to every altitude in a track cannot flip a sign, so the field elevation changes only *which samples are admitted to vote*, never how a vote is cast.

This is therefore a **coverage** result first. Any movement in accuracy is a second-order consequence of answering more flights, not of discriminating better between them. The report is written to keep those apart.

2 How a flight gets its aerodromes

Version 6 describes the three methods and the reasoning behind them; this section does not repeat that. What it adds is the part version 6 states only in prose — the **order** the pipeline does things in, and which datum each altitude gate is measured against — because the change this version makes is a change to that order.

Both methods answer two questions per track, separately: which aerodrome the flight departed from (ADEP) and which it arrived at (ADES). The shipped configuration takes departures from **endpoint** and arrivals from **trend**.

2.1 The steps

Every box in the diagrams below is one of these.

S1 — Read the track. One aircraft's state vectors for one flight, already split into a track by step 02 and cleaned by step 02a.

S2 — Smooth the altitude. A centred five-sample rolling mean over barometric altitude, computed over the *whole* track before any altitude cut. Smoothing after the cut would truncate the window exactly where altitude is changing fastest, which is the worst place to lose half of it.

S3 — Pre-filter on pressure altitude. A cheap, deliberately loose cut whose only job is to keep the aerodrome join affordable. Its ceiling is the configured cap *plus the highest field elevation in the reference*, so it cannot discard a sample that the exact test at S6 would have kept.

S4 — Join the aerodrome zones. Each sample is matched to candidate aerodromes through an H3 index at resolution 7, then narrowed to those within the detection radius by exact great-circle distance.

S5 — Attach the field elevation. Each candidate aerodrome's elevation, from OurAirports. A candidate whose elevation is unknown gets zero, which reduces the next step to the sea-level behaviour rather than dropping the aerodrome.

S6 — Cut on altitude. *This is the step this version changes.* Samples above the ceiling are excluded from the vote. **trend** now measures that ceiling **above field elevation**; before this version it was a flight level.

S7 — Vote. Each surviving sample votes on the sign of its smoothed altitude change: rising for a departure, falling for an arrival.

S8 — Decide, or abstain. If one direction leads the other by more than the margin, the track is departing or arriving. Otherwise **trend** says nothing.

E1 — Take the endpoint fix. **endpoint** reads one sample, not many: the track’s first fix for a departure, its last for an arrival.

E2 — Test the gates. The fix must be within the radius, and either flagged on the ground or no higher than the height gate **above field elevation**.

E3 — Rank the survivors. Nearest wins, on exact great-circle distance, plus a penalty for aerodromes without scheduled service so that a military or general-aviation field must be *clearly* nearest rather than merely nearest.

E4 — Or say where it came from. Where the gates fail and the fix sits at the edge of the observed area, **endpoint** reports **out-of-area** rather than guessing. Precedence is aerodrome first, border second: Ponta Delgada lies about 8 NM inside the western edge, and the other order labelled its departures out-of-area with the aircraft still on the runway.

2.2 trend, arrivals

Step	What it does
S1	Read the track’s state vectors
S2	Smooth baro_altitude over a centred 5-sample window, before any cut
S3	Pre-filter on pressure altitude at the cap plus the highest field elevation
S4	Join the H3 aerodrome zones at resolution 7, then cut on exact distance
S5	Attach the candidate aerodrome’s field elevation; unknown becomes zero
S6	Cut on height above field elevation — the step this version changes
S7	Each surviving sample votes on the sign of its smoothed altitude change
S8	If descending votes lead by the margin the track is arriving; else <i>undetermined</i>
E3	Rank survivors on distance plus the scheduled-service penalty; nearest becomes ADES

2.3 trend, departures

The same machinery reading the same samples; only the direction of the winning vote differs. **trend** has **no out-of-area concept** — it names an aerodrome or says nothing — which is why the shipped configuration does not use it for departures.

Identical to the arrival table above, with one difference: at **S8** the track is a departure when *climbing* votes lead by the margin. Failing that, **trend** returns *undetermined* — it has no out-of-area answer available to it.

2.4 endpoint, departures

Step	What it does
S1	Read the track's state vectors
E1	Take the first fix — one sample, not many
S4	Find candidate aerodromes within the radius
S5	Attach each candidate's field elevation
E2	Gate: within the radius and either on the ground or no higher than the gate above field elevation
E3	Passing: rank on distance plus the scheduled-service penalty; nearest becomes ADEP
E4	Failing, with the fix at the edge of the observed area: out-of-area . Otherwise <i>undetermined</i>

Precedence is aerodrome first, border second — the other order labelled Ponta Delgada's departures out-of-area with the aircraft still on the runway.

2.5 endpoint, arrivals

Identical to the departure table above, with one difference: at **E1** the method reads the track's **last** fix rather than its first.

2.6 Where the two datums part company

Only **S6** and **E2** read an altitude, and until this version they read it against different references:

Test	Threshold	Datum, before v6.1	Datum, from v6.1
trend S6	trend_max_height_ft	flight level (sea level)	above field elevation
endpoint E2	endpoint_height_ft	above field elevation	above field elevation
step 04 ground phase	phase_ground_ceiling_ft	above field elevation	above field elevation
nearest	<i>none</i>	—	—

nearest applies no altitude condition at all, on any datum. That is deliberate: it is the naive **traffic-faithful** baseline the other methods are scored against, and giving it a gate would defeat its purpose.

! A flight level is not a height

FL60 is not the same cut as 6,000 ft. The pipeline derives the flight level with an integer cast, so `flight_level <= 60` admits everything below **6,100 ft**. The datum comparison in this report therefore holds the ceiling at 6,100 ft above field, not 6,000 — otherwise it would move the ceiling and the datum at the same time and could not attribute the result to either.

3 Results

 Not yet measured

The arms defined in `regenerate_v61.py` have not been run against this checkout, so this section is deliberately empty rather than populated with numbers carried over from another study.